



A Primer to Human Threading

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ABSTRACT

Human Threading™ is new approach in developing innovative computing technologies. It uses novel physiologic combinations to measure the human brain and body in an effort to create greater efficiency among human and machine. A divergent group of measurement techniques are employed in this process including EEG, NIRS, fMRI, GSR, Saccade testing and more. An overarching goal in this area of study will be the development of simultaneous, multi-modal operating platforms providing efficiency and inter-changeability. As an example this paper provides a comparison of efficiency between two email systems. The first system is a standard Microsoft Outlook-based email application using Microsoft XP. The next is a voice-based email system created at the Human Threading™ research laboratory. EEG and time trials were performed on ten participants with a total of 30 trials. Voice-based email was shown to be 220% faster and use statistically less cognitive load on participants prefrontal cortex. This paper shows initial results in support of Human Threading™ and the potential power behind efficient, modal specific operating platforms.

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1. Introduction

Human Threading™ is an initiative focused on optimizing human and machine interaction. It is a transdisciplinary field joining together observations in human cognition, neuroanatomical structures, finite state machines, and their associated relationships. The central motivation behind this study should be an identification of physiological inefficiencies between human and machine. As a result of this observation, new artifact creation or enhancement of existing artifacts should be realized. The scope of these observations has vast possibilities from personal computing to the use of embedded micro/nano based systems and beyond. An overarching theory behind Human Threading™ is design science (Walls, Widmeyer, & El Sawy, 1992) which may garner an unlimited array of artifact creation should they follow particular guiding principles. The notion of technological design (or redesign) may be developed based on current research in cognitive neuroscience, computer science, electrical engineering, psychology, mathematics, and more. Deliverables in Human Threading™ should consist of literature, detailed technological schematics, or technological artifacts.

Human Threading™ as a model may be viewed as three linear stages that recursively: (1) observe a given human's interaction with a machine, (2) design a more efficient way of interacting (if necessary), and (3) provide an output of this new relationship with the least cost and greatest operational efficiency between humans and computing appliances.

In stage one; the Human Threading™ process analyzes the temporal and behavioral response of a human when being stimulated by a given technology. Specifically, this stage will measure what neural cortical regions of a human are being stimulated by a given technology along with other physiological measurements. These observations may be conducted through single tests such as measurement in gross neural firings, functional Magnetic Resonance Imaging (fMRI), electroencephalogram testing (EEG), Transcranial Doppler Sonography tests, infrared spectral analysis, interaction based time complexities, Galvanic Skin Response, and more. However, most Human Threading™ research projects will combine several of these observational methods in an effort to better refine human physiological dynamics with a given technology. The next part of this stage seeks to determine whether this given technology is using the brain's resources (cortical regions) efficiently. To define efficiency, an evaluation between neural stimulation in the human and the technologies main objective will be analyzed. Additionally, a focus on what other activities the human is doing and how else this may be interfering with them physiologically (competing for similar physical and neural resources) may also be looked at.

In stage two of the Human Threading™ process, a focus on the design or redesign of an artifact is developed. This design stage has one central objective: to create a more efficient cooperation between human and machine. The design stage may concentrate on the development of an entirely new technology whose goal is to achieve the original objective in a more efficient manner. Motivation behind new artifact development will likely be a result of private commercial off-the-shelf products limiting application programming interface extensibility. Another motivation behind

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new artifact creation could be geared toward a physiological isolation of modalities. Furthermore, this stage may bypass a tangible design method altogether and focus on clinical in vivo testing (embedding an artifact).

The final stage of Human Threading™ will produce a desired output which must be repeatable and consistent. Based on prior stages, the output may be a new design artifact, a blueprint, or a more efficient technology. No matter the output, it must fulfill its obligation to enhance efficiency among humans and computers. Furthermore, the output artifact delivered by a given Human Threading™ study must be testable and repeatable beyond the examination vehicles used any particular study.

Human Threading™ may be looked upon as fresh step in the future design of information systems and technology. Twenty years ago its existence would be impossible. That is, humans were still developing ways to make basic computing devices which could provide practicality and usefulness to the masses (personal computers). Commercial off-the-shelf technologies did not exist or were exorbitantly expensive, and sufficient, reliable protocols were in early development. Furthermore, the price of testing human physiology dwarfs today's stable, low cost measurement tools. Nonetheless, with today's maturity in micro processing, memory, knowledge base of neuroscience, and the proliferation in open standards, research and development in human and machine efficiencies are primed. Human Threading™ looks to reshape computing architectures and landscapes through scientific testing and technological invention (design science theory). The basis behind this study looks to dissipate an all-in-one computing device and allow a human to more naturally and efficiently interact with computing agents. The strength behind compartmentalizing human and computer relationships should allow for new technologies to be designed with neural efficiency at its core. This neural efficiency should work alongside current technologies as well as emerging ones such as micro and nano-based systems. As a partner in human–computer interaction (HCI) research, Human Threading™ will provide a knowledge base of design theory grounded in scientific observation.

Though Human Threading™ is focused on discovering relational efficiencies among humans and machines. There are some natural derivatives which will be observed as a by-product through this study. Namely, sensory modalities and associated cognitive enchantment will be documented while investigating technologies. This means a Human Threading™ study may yield additional data beyond human and machine efficiencies which may encompass what participants like or dislike about a particular technology (and what sense it is using: smell, touch, taste, sight, sound). This notion of cognitive enchantment may both be measured by a participant's verbal response (subjective measurement) as well as data found in brain imaging. Through this derivative alone, Human Threading™ will provide commercial developers as well as researchers with granular physiologic insight which may directly improve technological development across multiple domains.

Indeed a primary artifact expected to surface from the formulation of Human Threading™ may be new and efficient ways of human multitasking. This will be a long-term and ongoing investigation as research cultivates around efficient human patterns. Current studies show there to be significant limitations among humans and multi tasking (Lohr, 2007; Rubinstein, Meyer, & Evans, 2001). However, it is to be examined through extended Human Threading™ studies whether adding layers of such neural efficient patterns (similar to automated robotic agents) may generate more successful multitasking results (Agmon & Peleg, 2004; Hongwang, Yufan, & Cuiping, 2006). This type of hands-off agent direction may greatly enhance human life and present itself as a future avenue of commercial development.

Several additional by-products are expected to come from this transdisciplinary study. However, a final upshot worth inspecting is the observed consequences of technological design based on neural efficiencies. Early questions at this point focus on the potential of artifacts to be created with a *zero trained learning curve* (allowing a human to immediately start using an artifact with little or no training). Also, what would be the effects of optimizing sensory modalities and the human performance around such artifacts?

2. Theory empowering Human Threading™

Design science theory as a rigorous research guideline (Hevner & et al., 2004) and practical methodology (Peppers et al., 2007) has been leveraged as the underling theory behind Human Threading™. Design science (DS) was introduced into information systems and technology as a fundamental value based on stratified concepts of theory building and science of the artificial (Dubin, 1969; Simon, 1969). It is understood in theory as a process which satisfies real world problems with logical and creative answers (Nunamaker & Chen, 1990). The applicable DS methodology in Human Threading™ will be carried out using the design science research methodology (DSRM) Peppers et al., 2007. This particular vehicle most efficiently fits Human Threading™ research as it encompasses previous design science work on theory, (Nunamaker & Chen, 1990; Walls et al., 1992) applied engineering, (Archer, 1979; Eekels, 1991) and identification (Rossi & Chatterjee, 2003) with an emphasis on the Hevner et al. guidelines.

The DSRM was developed with six activities which use recursion to bridge from one to another. The power behind such a model is the inherent flexibility with regard to each stage in the framework. Depending on the phenomena in a given Human Threading™ experiment, an entry point for design could fall into any step and follow the process iteration. These activities include: problem identification, defining objectives, design and development, demonstration of the artifact, evaluation, and communication.

3. Brain measuring and cognitive load

Human Threading™ focuses on calculating and analyzing neural stimuli through physiologic measurement techniques. As a result, a more likely approach to the design science research methodology will follow a nominal sequence as shown below moving from left to right (Fig. 1). The problem identification step will often times be novel and undiscovered. Primary motivation and justification behind this step may concentrate on an effort to reduce time completion or discovery of high cognitive load.

Cognitive load may be understood as the mental resources an individual exercises while solving problems or completing a task (Oviatt, 2006). There has been a tremendous amount of research devoted to accurately deriving cognitive load from humans during task completion (Adey, Walter, & Hendrix, 1961; Michels et al., 2010). Cognitive load theory (Patrick, 1979) gave rise to investigation in educational instruction and psychological fatigue of the human brain during the 1970s. Later, this theory produced meaningful (and practical) discovery with the delineation of working memory, (Sweller, 1988) and cognitive overload (Chandler & Sweller, 1992). Cognitive overload may be understood as the proliferation of data in short term memory which at a certain point overwhelms the individual.

As brain imaging and measurement technologies have flourished in recent years, so has the identification of key factors pointing to working memory and cognitive load. Electroencephalograph is an excellent example of how modern technology has provided cheap, fast indicators through post-synaptic potential measurement (Brunken, Plass, & Leutner, 2003; Curtis, 2003). There are

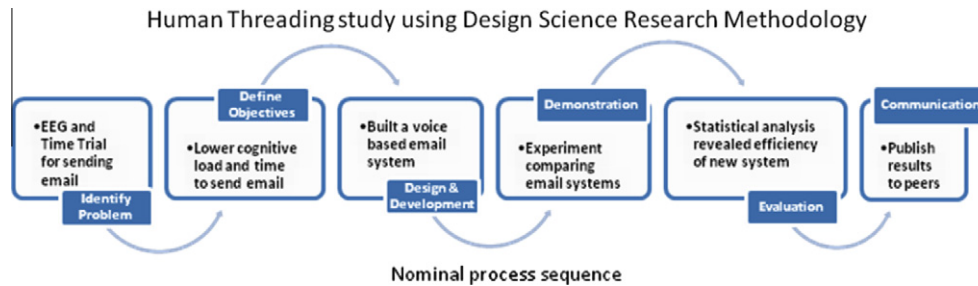


Fig. 1. The Human Threading™ research example demonstrates how it flows from start to finish using the Design Science Research Model.

both competing and complimentary brain-measuring techniques that researchers commonly use today from fMRI and EEG to infrared tomography and more (Akgül, Akin, & Sankur, 2006; Cordes & et al., 2001; White & et al., 2009). One commonly used and accepted measurement of cognitive load is discovering event related potentials (ERP's) via electroencephalography (Shenoy & Tan, 2008).

ERP's are brain phenomena which produce multi-dimensional amplitudes (both positive and negative polarity) through time locked control of a given stimulus (Luck, 2005; Zani & Proverbio, 2003). Researchers often times introduce participants to a stimulus and look for unique potentials milliseconds after the stimulus has been presented. Human Threading™ will make use of the P300 and other ERP phenomena in future studies by gaining a baseline of a participant then introducing them to a given stimulus (technology in this case). The resultant stimulus will be analyzed milliseconds after the introduction (Balconi & Lucchiari, 2007) depending on phenomena.

Aside from P300, advancements in other neurological research areas have provided an array of measurement processes for studying the brain. Each specific measurement mechanism or process has its own particular set of advantages while simultaneously may offer some disadvantages (Lee & Tan, 2006). Some of today's more popular methods include Magneto-encephalography, Single Photon Emission Computerized Tomography, Positron Emission Tomography, Functional Magnetic Resonance Imaging, Event-Related Optical Signal/Functional, and Functional Near-Infrared Imaging. As discussed, a common brain measurement methodology incorporated in Human Threading™ studies will be the EEG. This measurement system provides researchers with tremendous temporal resolution (Alan et al., 1998; Ferrez & Millán, 2005; Gerč & Jaušvec, 1999; Gevins & et al., 1997; Holm & et al., 2009), and is a very important tool when examining human and computer interaction.

EEG research is a non-invasive procedure whereby electrodes placed on top of the scalp read neural activity (post-synaptic action potentials) on the outer most portion of the brain (neocortex) Nunez & Srinivasan, 2007. Post-synaptic action potential assessment via EEG has made significant contributions to Human Computer Interfaces and Brain Computer Interfacing as an accurate method in studying specific brain reaction to stimulus (Shenoy & Tan, 2008). More specifically, EEG has proven itself as a reliable process in observing potentials through several modes of spectral phenomena beyond ERP's. For example measuring participant's pre-frontal theta activity alongside parietal alpha potentials has also been proven a recurring gauge of cognitive load (Gevins et al., 1997; Holm et al., 2009; Sauseng & et al., 2005).

However, to its detriment, many studies reveal there is no silver bullet method to EEG and other brain measurement discovery (Alan et al., 1998; Hruby & Marsalek, 2003). Brain measurement is still an imperfect science which offers what some researchers consider crude representations of neuronal activity (Ferrez & Millán, 2005). As a result many Human Threading™ investigations will

incorporate other measurement techniques alongside EEG, fMRI, and infrared spectroscopy in an effort to realize a consensus. These additional techniques may include galvanic skin response measurements, (McCleary, 1950; Paintal, 1951) subjective workload questionnaires such as the NASA TLX (Botzer & Meyer, 2008; Park, Harada, & Igarashi, 2006), electrocardiogram (Cao, Theune, & Nijholt, 2008), saccade tests (Lee, Rohrer, & Sparks, 1988) and more.

4. Human Threading™ in motion

To show an example, a recent Human Threading™ research project investigated the effects email had on humans in relation to their workplace. Previous research estimates email to save companies 326 h per employee every year (\$9000 net savings per employee) over the use of telephone (Korzeniowski, 2000). However, studies focusing on media richness theory counter this with evidence illustrating how communication decomposes as it deviates from voice (Brunelle & Lapierre, 2008; Media richness theory, 2010; Te'eni, 2001). That is to say it is more natural (and better understood by the recipient) to speak a message rather than typing it. As a result, researchers at Human Threading™ created a novel vestibular based system engineered to target speech and audition. This voice-based email system was invented to isolate speech and mitigate simultaneous reliance on additional modalities such as sight and touch (ergo keyboard, mouse, and LCD display).

Thus, the scope of this Human Threading™ study compared a standard Microsoft Outlook-based email system using Microsoft XP with a voice-based email system to determine two criteria. First, was it faster for participants to send an email using the standard pc/Outlook system they currently used at their workplace rather than the voice based system? And next, were there any differences in cognitive load between the systems?

To progress through the DSRM in a nominal sequence, time and cognitive load results produced a problem centered initiation at research entry point 1 in the DSRM. That is to say, researchers found current email systems time complexities unacceptable and hypothesized to reject its cognitive load values at the prefrontal and parietal cortices (Fz and Pz in the international 10–20 system). The investigation for a solution (and the objectives thereof) was established to lower cognitive load while speeding up the time to send an email (activity #2 in DSRM). This improvement could save businesses tremendous amounts of capital if hypothesis H_0 were rejected (standard email system). A voice based artifact was developed in light of these prior two steps (activity #3) and demonstrated (activity #4) with 10 participants to meet the objectives set in step 2. Finally, after a successful evaluation via EEG and time trial (activity #5) this data has been submitted as communication literature (activity #6).

4.1. Participants

All applicants were required to have a working knowledge of Microsoft Windows and Outlook 2007 through business practice.

Informed consent was agreed to by all participants and each EEG session lasted approximately one and a half hours. Prior to each experiment participants were given a template consisting of the email they were to send (To, Subject, and Message data). This template required users to send the email to a generic hotmail account with a subject of 'Antony'. The message data was two sentences used from *The Tragedy of Julius Caesar* by William Shakespeare, 1599. A brief walkthrough of each email task ensued in an effort to make sure participants understood their responsibilities.

4.2. EEG Measurements

EEG was recorded at all channels simultaneously using an Atlantis 4 × 4 with gold plated electrodes. Based on previous work, a monopolar montage was applied computing spectrograms on the International 10–20 system derivations of Fz and Pz with a corresponding reference on A1 and A2, respectively (Alan et al., 1998; Brookings, Wilson, & Swain, 1996; Gevins et al., 1997; Holm et al., 2009; Sauseng et al., 2005). Ground was set at Cz with a resistance rate of less than 5 k Ω . EEG sampling rate was set to 256 sps using Brainmaster 3.4 software with a peak to peak amplitude scale. Spectral power values set for theta (4–8 Hz) and alpha (8–12 Hz) was evaluated using a Fast Fourier transformation and all data was analyzed by means of 1 s epochs. Artifact was corrected for ocular and muscular interference through a built-in application analysis utility as well as manual inspection.

5. Results

A matched set hypothesis test for paired samples were assessed under the following assumption: $H_0: \mu_d \leq 0$ while $H_1: \mu_d > 0$. The t -test was carried out as follows $t = \bar{d} - \mu_d / s_d$ where $s_d = s / \sqrt{N}$. Both frontal theta and parietal alpha were investigated with degrees of freedom (df) = 9 and confidence limits set to 95% (critical value for $\alpha = .05$) where the t -distribution = 2.262. With a calculated frontal theta value of 3.9966, the decision to reject H_0 is statistically significant as it surpasses the critical value 2.262. However, parietal alpha does not reach the critical value for rejection ($t = -2.6265$).

With a sample size of 10 and 60 tasks (30 competing between Outlook and voice), we can say with 95% confidence that the average frontal theta frequency modulates between 19.351 and 23.427 while using Outlook. However with the same confidence factor, speaking the email through the voice based system averages 19.648–16.052. Similarly, we can say with 95% confidence that the average parietal alpha frequency lies between 8.538 and 6.868 while using Outlook. Speaking the email raised parietal alpha spectrum averages from 9.94 to 7.634 (see Figs 2 and 3).

A two tailed t -test of equal variance was conducted to measure time with s_p^2 representing a pooled estimate. With df = 18, $H_0: \mu_1 - \mu_2 = 0$ and $H_1: \mu_1 - \mu_2 \neq 0$. Confidence limits were set to 47.5 on each tail ($\alpha = .05$) and the critical value was established at 2.10. Outlook analytics were estimated having a mean of 66.1667 and standard deviation of 29.4619. Voice based mail had a mean of 30.0667 and standard deviation of 4.697. As a result of the t -test, $s_p^2 \sigma = 21.096$ and resultant test statistic = 3.83. As the calculated value of the test statistic is greater than our critical value, H_0 must be rejected.

6. Discussion and conclusion

Human Threading™ is process which measures human interaction with machines for future development of simultaneous multimodal operating platforms. It aims to discover inefficient relationships and bridges these obstructions with artifacts. As a transdisci-

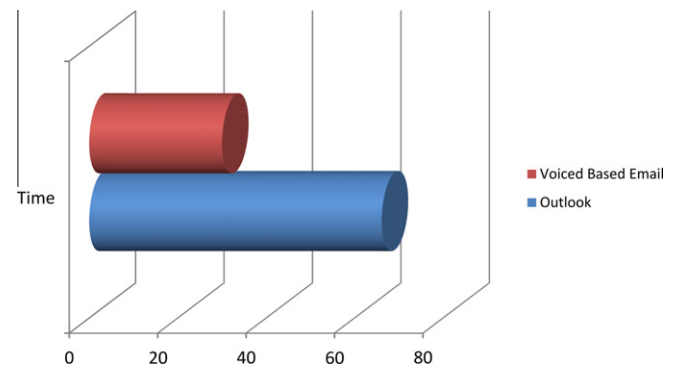


Fig. 2. Average time spent sending email. Outlook on average takes more than twice the time to send an email than the unfamiliar voice based system. Outlook Mean = 66.167 while voice-based email Mean = 30.067.

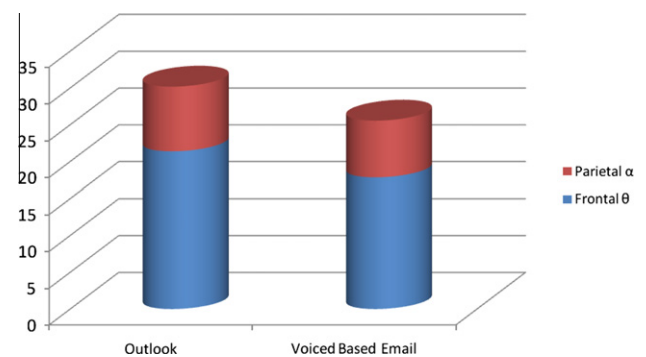


Fig. 3. Cognitive load measured in the prefrontal cortex with a focus on theta was significantly higher while using Outlook. Parietal alpha also increased while using Outlook, however albeit a much smaller difference.

plinary, applied system drawing on recursion and rooted in design theory this subject matter will undoubtedly generate novel technologies. Human Threading™ discovery offers researchers and designers an unprecedented window into human tendencies in relation to machine usage. It involves experts in many different areas of science and engineering to complete its mission accurately. Furthermore, as human patterns of relational usage with machine are recorded using physiological analytics, Human Threading™ hopes to discover methods and efficiencies currently unknown in HCI research.

The example presented in this paper is an exact case-in-point. Data has revealed several interesting findings which have direct implications on theory, development, and business practitioners. The original investigation sought to discern whether there was any utility gained by speaking an email rather than using our current approach of typing. Based on this study, the direct answer to our question points to yes. Participants put forth much less mental effort while speaking an email compared to using a standard personal computer and typing it. The prefrontal cortex almost unanimously showed a lower theta frequency with the voice based system compared to Outlook. However, with regard to the inverse effect of an increased parietal alpha, there was no statistical significance. Both frontal theta and parietal alpha seemed to decrease in frequency while speaking an email rather than typing it. What seemed most surprising about this result was participants had never before used the voice based system. Nonetheless, they seemed to reveal less mental effort in using it compared to a competing system they use at work every day. This result could mean many things from a neurological perspective. Perhaps the reason participants displayed smaller amplitude when speaking their

email correlates to less mental effort using working memory and task switching (Ramnani & Owen, 2004).

The second metric used in measuring business utility also showed a statistical significance in favor of speaking an email. It was 220% faster on average for a participant to speak the email rather than type it. Again, this mean result was a consequence of participants first interaction with a voice-based email system. Considering the favorable outcome on both dependant variables, a new question is posed: after using a voice-based email platform for a considerable amount of time, do participants spend even less time and cognitive load than revealed in this study? Nonetheless, the findings in this investigation point toward a greater utility earned by firms who adopt a speech based email system. To be specific, currently companies who enjoy a saving of 326 h a year (or approximately \$9000 per employee) using a standard email platform could save 717 h or about \$20,000 through use of a voice based platform.

Another important aspect this research unlocks is a call to developers who produce software and hardware systems. Should this data be strengthened by future research with similar results, a voice-based email system may be highly sought after (from a financial viewpoint alone). Fabrication of such an artifact from the statistics presented in this study certainly may call for future development. However, this is not to impute such a system would have immediate characteristics for adoption. There are limitations to the development of this study and its technologies. For example, designers need to be cognizant of the immediate shortcomings associated with adopting a spoken language system. It may be disadvantageous to embrace such a system if employees work close to each other or if sound is a factor in the workplace. The voice-based email system used in this experiment would not be suitable in a real world production environment. It relies heavily on sparse kernel machines and only tolerates low background noise while sending an email. Another limitation Human Threading™ will most certainly be faced with are individual differences. The notion that future experiments encompass entire populations is highly unlikely. This research anticipates a contribution to knowledge from a gross prospective of participant data. Artifacts derived from the DSRM should be made practical for many people, but certainly not all.

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